daml.robertab.nl

Assignment 2: Text clustering

4–5 minutes

Introduction

In this assignment you will build clustering models on text data, specifically on reviews from the Internet Movie Database (IMDB).

- IMDB reviews is a text dataset available in the text2vec package.
- This dataset consists of 5000 movie reviews specifically selected for sentiment analysis.
- No single movie has more than 30 reviews.
- The sentiment of the reviews is also given as a binary variable, i.e. an IMDB rating < 5 results in a sentiment score of 0, and a rating >=7 results in a sentiment score of 1.

Your task is to cluster your text data, perform a comparison and write a report.

AI tools / ChatGPT

The skills assessed in this assignment include your capability to write code, to communicate clearly about this code and its results, and to understand the components that go into model comparison.

Do not use AI tools such as ChatGPT for any part of this assignment. Also do not use AutoML systems.

- Apply at least one text preprocessing step after loading the dataset.
- Select a text representation method (e.g. a document-term matrix, word2vec) to represent your text column.
- Choose two or more appropriate clustering methods (e.g., K-Means, K-Medians, topic modeling, GMM, etc.). This can be any method, you don't have to limit yourself to the methods taught directly in this course.
- Apply all these methods to the IMDB dataset with 5 and 10 clusters and evaluate the models. This can be done using internal validation (measures such as DB-index, Silhouette coefficient, etc.) and/or external validation (e.g. using the sentiment labels).
- Produce a written report to communicate your work.

Written report

Your written report should contain the following elements. You can use this template for your report. Download the `.qmd` source here.

YAML header

when downloading the `.qmd` source, the header of the file gets lost. You can add the following header to the top of your file (include the dashes) to compile the file to a nicer `.html` file.

```
---
title: "Text clustering"
author:
  - Author One
  - Author Two
  - Author Three
```

```
date: last-modified
format:
  html:
    toc: true
    self-contained: true
    code-fold: true
    df-print: kable
---
```

- Data description & exploration. Describe the data and use a visualization to support your story. (approx. one or two paragraphs)
- Describe text pre-processing steps you have used (approx. one or two paragraphs)
- Briefly describe your text representation method. (approx. one or two paragraphs)
- Briefly describe which models you compare to perform clustering. (approx. two or three paragraphs)
- Describe how you compare the methods and why. (approx. two or three paragraphs)
- Write down what each team member contributed to the project.

Hand-in

Create a folder with:

- Your report (.html or .pdf)
- The qmd or ipynb file that generates your report
- All the resources needed to generate the report from the file
- Any additional code files you have used

Zip this folder and upload it to Brightspace.

Computational reproducibility

This folder needs to be computationally reproducible. That is: if we download the folder, open the .qmd file and press “render”, the report is generated without error (potentially after installing the necessary packages). Make sure to check this (i.e., try it on different computers).

Grading

Your grade will be based on the following components:

- Code quality: Including legibility, consistent style, computational reproducibility, comments. (20%)
- Content: Whether all required components of the written report are complete, correct, and concise. (80%)